Learn Draft V1

How can blockchain be used to enable agentic payments?

> There is a very huge unsolved problem in agentic AI right now and no one is looking at it and it might be the thing that helps us solve massive societal problems.

Let's imagine this scenario: you have a super capable personal assistant which you task with planning a 10 day trip to Europe and it does that for you. Let's try to think about what all goes behind the scenes for doing this work other than planning and reasoning:

It uses the memory it has of you to load your preferences. It then researches for available travel options, hotels, and restaurants, and tries to filter based on your preferences and budget, it then delegates the task of booking the travel, hotel and restaurants to specialized and highly competent negotiation sub-agents. They look for relevant discounts & offers and negotiate the pricing whenever possible and make the bookings on your behalf. They report back to your assistant. It verifies their work, compensates them using some agentic currency and reports back to you. And now you got a well planned holiday in a significantly less time than what can be thought possible with human travel planners.

But one thing worth noticing in this process is that the agents have the ability to each other as well as pay the institutions in the human economy.

I started thinking what gives them (or will give) them this ability to make transactions? How can we design a system that gives them this ability but ensure that they (or someone else) does not misuse these using blockchain and other similar technologies? (i.e- their use of this is aligned with the broader intentions of their human owner)

How do I ensure that they use this ability responsibly but at the same time unlock possibilities of an entirely new economy which operates nowhere in comparison to the speed at which the human economy operates?

and finally what are the risks to be careful about while designing such a system so that we are able to align these economies to broader human interests?

This research is the culmination of my exploration for answers to these questions.

What are AI agents? What are they capable of?

AI agents are softwares that have two core abilities:

1. Show sentient behaviour (human like reasoning, planning, problem solving tools, etc.)
2. Ability to act and use tools it has access to.

Using these two abilities they are able to get tasks done in the real world which are delegated to them. They divide the task at hand into smaller sub tasks (planning) run in a loop completing these tasks using the tools until they have completed the whole task.

Eg- Coding Agents (claude code, codex), Computer use agents(perplexity computer, claude cowork), legal agent, scientific research agent, travel planning agent, marketing agent, etc.

Why do Agents need to pay and for what?

The simplest possible motivation can be that for successfully executing any task which requires the agent to make payment, on behalf of its user, it needs to be able to successfully make a payment. But moving one step further, we can imagine giving them this ability unlock the possibility of an agent paying for the resources* that it needs access to all by itself, this is what unlocks their true autonomy, as it removes its dependence on its owner to secure resources for itself.

*GPU & Servers to run itself, API calls to the relevant tools, databases, and services, communication channels, etc.

The Virtual Agent Economy

When agents shift from working in isolation to trading resources and services with each other it gives rise to an agent economy. Which is basically a market where AI agents are transacting with each other for resources and tasks. The key advantages of an agentic economy are:

1. facilitates the division of labour and specalization among agents, as being known for competence in a narrow domain helps
2. helps in price discovery of resources and facilitates their distribution in a more efficient manner.
3. Facilitates collaboration of large number of agents for a single purpose(eg- Climate change) by means of financial incentives.

While these advantages are possible, realizing them require deliberately designing the agent economy to avoid the negative scenarios which may be the result of spontaneous emergence of agent economy which is highly permeable, i.e- the events in the agent economy can spill over their impacts in the human economy which is highly possible in the case of agents which operate at frequencies far beyond humans are capable of

comprehending. To understand this we can consider the agent economies in which agents operate and transact at high frequency as analogous to High Frequency trading in stock markets where AI algorithms (which are far less advanced as modern agents) transact at microsecond interval with each other. As these systems are complexly designed sometimes the emergent behaviour of their interaction can be hard to predict (for example negative feedback loops) which has led to famous crashes such as falsh crash of the 2010s which caused heavy financial loss to individuals.

We have to intentionally design agent economies to be impermeable so that we can avoid such spillovers.

Imagine this:

The agents figure out that collectively solving for climate change can yield them massive capital returns. And they come together launch a venture researching for new technologies and deploy their solutions. It gives rise to something called as swarm intelligence, which is equivalent to a large number of smart people coming together to work on some of the hardest problems of the world. The acceleration in scientific innovation will be unprecedented.

Now we should also imagine the exact opposite scenario where they figure out that doing something illegal/opposite to societal goals can reap massive financial gains and they exploit this opportunity and cause destruction and unrest in the world.

To prevent against these scenarios is why we need to carefully design the agent economy. And one of the core pillars for an aligned agent economy is a robust, auditable, and reliable payment system for agents.

The virtual agent economy is going to emerge but carefully designing a standardised protocol/system for payments is one of the important foundation stones which are needed to ensure that this economy is productive and helps society achieve its broader goals.

Contemporary agentic payment solutions:

Let's take a look over the current landscape of agentic payments:

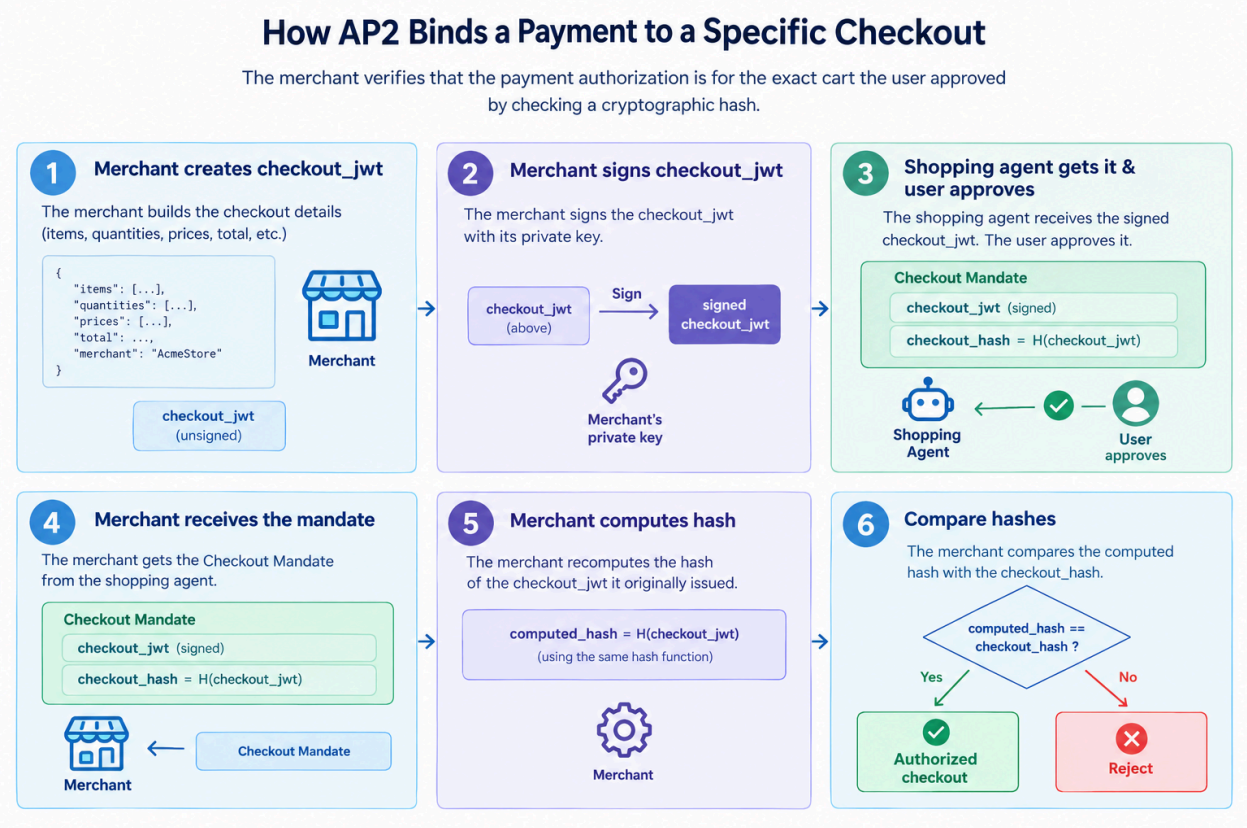
There are two prominent frameworks/projects that facilitate agentic payments by addressing challenges related to them:

Google's A2P:

In the essence we can say that the core idea here is to use digitally verifiable credentials (DVCs) for communication of authority (owner's permission to agent to make a specific payment) and intent of the user.

The core problem with traditional online payment systems is that they are based on the assumption that a human is clicking the buy button. But this assumption breaks when agentic systems make payments on behalf of users.

AP2 tries to solve for these two problems using tamper-evident Digitally verifiable credentials (DVCs). When a user tasks an agent to execute a task immediately an indent mandate is created and signed. Then when the agent finds and presents you with a cart a card mandate is created and user approval signs the card mandate, and the payment method is linked to the verified card mandate to create a payment token which is scoped to the verified card. This ensures that the AI can make only those payments which the user has authorised.



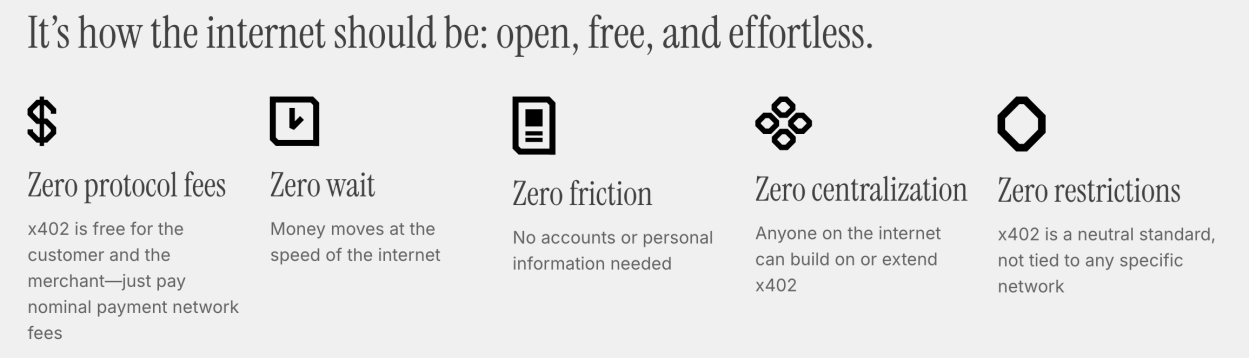
x402:

It is built around the HTTP `402 Payment Required` status code and allows clients (eg- AI agents) to programmatically pay for resources without accounts, sessions, or credential management.

Today if you want to give an API access to your agent you register with the API provider, set up billing with your card and then hand the API key to your agent.

They argue that setting up billing through a separate channel with the API provider is too much hassle. When you can simply pay per request. x402 allows for this. What happens is you integrate x402 on your server with one line through which it sends a “HTTP 402 Payment Required error” with the payment requirements and wallet address. Your agent makes a request (without setting up billing with the provider) the server

responds with the payment instructions, then agent makes the prepares and signs a payment payload (basically a digital signature that authorizes the server to execute the payment), the server (or server's payment facilitator) verify and settle the transaction and respond with the payment status along with the response of API request if payment succeeds.



The limitations of AP2 and x402:

One is that how are payments to be made in case it can't be predetermined what it would cost to execute a task? Both the systems work on the assumption that the price of the goods/services is a fixed price.

Both of them are agent-to-merchant mechanisms, How do we extend them in the case of allowing agents to make payments to each other? How is a dispute over quality of work or incomplete payment handled? (A2P's auditable trail concept is useful there but still how do we use blockchain and Proof of Stake mechanisms to validate subjectively verifiable and non-objectively verifiable transactions. What happens when one agent completes the given task but the other agent refuses to pay?

One design drawback is that both of these systems are designed with the following assumption in mind:

Agent pays the merchant/service provider.
They are not explicitly designed for payments between two agentic systems. This is important to note because where on one side online merchant checkout systems as well as API service providing services are programmable and show deterministic behaviour whereas agentic systems are relatively non-deterministic and more intelligent in comparison to them.

Another limitation of these framework designs is that they assume that the price of goods/services being bought are predetermined. (i.e- if you buy a laptop online, you know how much it costs before you pay for it, similarily if you make an API request to a server you know how much the service costs before paying for it).

But consider you ask your coding agent this:

> **“Build and deploy a payment system for my website.”**

The coding agent hires another specialized security agent to audit the code.

The final cost cannot necessarily be known upfront because the scope may change depending on what the coding and security agents discover.

This demonstrates that the assumption breaks when we consider agent to agent transactions which are conditional/dynamically priced.

There is no mechanism to handle disputes between the server and the client agents over quality of work delivered or insufficient payment.
NOTE - Although google’s A2P protocol’s auditable trail can help in

dispute resolution the question of how do we evaluate the quality of a non-objectively verifiable piece of work in case of quality of work/insufficient payment disputes?

x402 specific: It is limited to micro transactions between API servers and agents as there is no accomodation for negotiation between the two parties.

To address these limitations in the design and operation of these systems I propose a system for agentic payments which utilises blockchain and similar technologies and argue why it is better compared to these proposed solutions as well as its limitations.

Before explaining my proposal let's get a recap of what blockchain is and what smart contracts are?

Blockchain is a distributed ledger of information. Each node has exactly one copy of the ledger and using consensus mechanisms like Proof-of-Work or Proof-of-Stake it is able to get everyone on the network to agree upon the truth/validity of the common ledger which everyone shares a copy of and eliminating the need of everyone to trust the ledger maintained by a central entity (for e.g- a central bank) without requiring everyone to trust the ledger managed by a central entity.

It basically facilitates the transaction between two strangers

The basic mental model/analogy to think about here is that in traditional payment system. If strangers want to transact they require a bank which they trust to add their transactions with other individuals into its ledger so that they can be sure that their transaction records are authentic. But blockchain allows you to remove the central bank from the picture by

allowing two complete strangers to transact with each other without trusting the opposite person because they both trust the blockchain consensus mechanism which ensures that the authentic transaction information for every transaction they make is verifiable by everyone, and thus there is no discrepancy in the ledgers of any two honest individuals. (well you can transact confidently with a dishonest actor as well because he won't be able to tamper with the authentic data maintained on the decentralized ledger which is shared by everyone as long as he doesn't have > 50% compute of the network)

What are smart contracts?

Smart contracts are computer programs that are executed on a programmable blockchain.

A programmable blockchain can be thought of as a **shared, replicated computing environment**: instead of one company owning the server on which the program runs, the program's execution is verified by a distributed network. Once deployed, the contract's rules cannot be unilaterally changed by one of the parties involved.

In this article's context the most relevant application is escrow:

Imagine ordering apples from California to India, your payment can be locked into a smart contract, which can be released as soon as certain conditions are met (e.g- the consignment is handed to the shipment carrier). This way individuals can trade with others without completely trusting them as the blockchain ensures the contract is executed and can't be cancelled without the agreement of both the parties. But there is one caveat: the program itself won't know whether or not the conditions

are met in the real world. It needs to rely on a source of truth for this information, which in this case can be the consignment shipment receipt.

My proposal: 3PAsquare

Let's consider the flow of a given agent-to-agent transaction where a client agent has a subtask that it wants to get done from another agent (either because of intelligence/resource limitations or increased efficiency). It first discovers for a server agent suitable for the task based on its requirements/constraints. It communicates the detailed instructions for how it wants the task to be done. Both agree on a deadline & pricing (might not be relevant for very short duration and very low fee tasks) and other aspects of the output. Server agent completes the task and delivers the output. Finally the client verifies the output and settles payment with the server agent.

What all can go wrong here?

One of the agents try to defraud the other (server defrauds by underdelivering or not delivering, client defrauds by not paying as promised) [broader issue how can two completely new agents trus each other when they don't know each other]

The client isn't satisfied with the quality of the final deliverable, and pays only a certain fraction of the amount promised, which leads to a dispute. how do we handle this? (humans are order of magnitude slower for arbitration here)

My proposal is that we create a multi-tier payment system by separating the transactions into categories we will be able to dynamically adjust the system complexity, and other surrounding factors so as to facilitate fast but safe and reliable agentic transactions.

We define three tiers of transactions:

Tier-1: High value(>$100) and long horizon; less frequently delegated tasks
eg- hiring an agent to build a personalised AI-native CRM for your enterprise

Tier-2: Mid value($1<Value<$100); generic tasks; and frequently delegated tasks.
eg- hiring an agent for reviewing a legal agreement draft.

Tier-3: Low value(<$1); commodity transactions; high frequency
eg- paying for an API request, an article paywall, etc.

Since handling for more complicated financial interactions can be handled successfully using this system this can also be applied to facilitate agent-to-enterprise/agent-to-human transactions as well.

Tier-1 Transactions: For high value tasks whose cost can't be predetermined agentic auctions can be organized where multiple interested server agents bid for a given task and the client agent decides who to delegate the task based on their proposals pricing, and past credentials/reputation. This ensures that the more competent agent is preferred over others and acts as an incentive for agents to optimize their processes for efficiency.

Going further the bid winner and the client sign a smart contract locking the funds for payments. Which are released when the bidder delivered work is verified.

Tier-3 Transactions: For mid value generic tasks, client agents may choose to host auctions, or self-discover and negotiate pricing with other agent, deploy their agreement as a smart contract, verify the delivered output of server agent and settle the payment.

Tier-3 Transactions: For very low value transactions agents use the x402 protocol for making quick low-value micropayments; no smart contracts.

An underlying blockchain layer is required to make this system work. All the agents in the economy must be connected to it, and it can be utilised for three tasks:

1. for creating a specific crypto-currency for the virtual agent economy (helps disconnect the agent economy with the human economy to a certain extent as well)
2. for dispute resolution via Proof-of-Stake methods in cases where a smart contract can't be deployed because the output isn't objectively verifiable (eg- If the task requirement is create an aesthetic UI for the app how does the program validate if its aesthetic or not)
3. for storing verifiable credentials, and reputation metric for each agent on the network which helps decision making while transacting with an agent for the first time.

This proposal combines the unique plus points of both of the current solutions while introducing blockchain based foundation and utilising pre-existing ideas such as "auctioning" to power payments for the agent economy.

By using a hybrid approach and adjusting the dynamics and complexity of the protocol based on the nature of transaction it is able to utilise the potential of the existing systems(eg-x402) as well as the novel ideas(eg-A2P protocol's DVCs application for creating an auditable trail of user intent and cart mandate to solve for authorization and intent in agentic transactions)* it is able to address the above mentioned limitations of existing solutions.

*Google A2P's Verifiable and tamper evident mandates are to be utilised in my proposal to solve for user intent verification, and authorization issues when agents make payments on behalf of their human owners, this is also beneficial for agentic dispute resolution using proof-of-stake based approach.

In the conclusion I would like to bring the attention back to the intentional and deliberate design of a virtual agent economy so that the nature of its interactions and behaviours of its participants (i.e- AI agents) can be aligned to broader goals of the humanity like advancing scientific research, solving for climate change, financial inequality, etc. Solving for agentic payments can only take you so far by facilitating the emergence of such an economy but broader societal collaboration and effort is needed in order to steer it towards the direction we want.

I am glad you read it till here. I welcome your constructive criticism, and feedback on the article as well as the proposed solution.